

THE GREAT LEAP: RTS RESPONSE TO BOUSSO'S CLAIM REGARDING THE IMPOSSIBILITY OF A CYCLIC UNIVERSE

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ABSTRACT

Physicist Raphael Bousso argues, based on an extension of Penrose's theorem, that a cyclic universe is impossible and that the Big Bang singularity is inevitable. In VG6, we demonstrate that this conclusion holds only within the framework of standard quantum mechanics and classical cosmology. In RTS, the introduction of the **subquantum wall** and **nodal fragments** eludes Bousso's critique, transforming the singularity from an inevitable endpoint into a **transition gateway** through which the Network passes, preserving its memory and coherence. The episode provides a step-by-step argument and proposes a list of **observable imprints** of the Great Leap, which can be searched for in the cosmic microwave background, in the distribution of dark matter, and in low-frequency gravitational waves.

Keywords: Bousso, subquantum wall, nodal fragments, Great Leap, cosmic imprints, VG.

1. BOUSSO'S CRITIQUE – AN ANCHOR FOR REFLECTION

Raphael Bousso, from the University of California, Berkeley, has recently demonstrated that, within extended quantum mechanics, an initial singularity is inevitable, rendering any cyclic universe model impossible. His argument is based on a generalization of the second law of thermodynamics and on Penrose's theorem, which shows that under extreme conditions, space and time "break down" and the laws of physics cease to function.

In VG, we acknowledge the strength of this argument. Bousso is right – **within the limits of the standard model**. However, RTS adds an element that standard quantum mechanics does not take into account: the **subquantum wall**, a threshold structure that exists before and after the singularity and allows the transition between cycles without violating known laws of physics. Here is how.

2. WHAT IS THE SUBQUANTUM WALL – A CLARIFICATION FOR BOUSSO

To respond to Bouso, we must revisit what we established in VG5, but in a concentrated form, directly addressing his argument.

The subquantum wall is an **emergent property** of the Network, manifesting as a **threshold** field that surrounds the Network like an airtight alveolus, without fissures. It is not a physical object, but a **coupling structure** that exists in a state of "waiting" during expansion and is activated only at the transition between cycles.

What does this wall do?

- **Accumulates** energy and information that would otherwise be lost during the contraction phase – that is, when the Network re-winds.
- **Stores** the memory of the cycle in the form of **nodal fragments** – coupling structures that are not subject to ordinary spatio-temporal constraints. These are not particles, but **pure patterns** of the Network, which can be activated instantaneously at the opening of the new cycle.
- **Releases** the information back into the Network at the beginning of the new cycle, ensuring the **continuity of cosmic memory** beyond the singularity.

3. THE SINGULARITY AS A GATEWAY, NOT AN END

Bousso considers the singularity as an **endpoint**, a place where the laws of physics break down and from which there is no return. In RTS, this picture is correct for a Network without a subquantum wall. However, thanks to this structure, the singularity becomes a **transition gateway** – a moment of maximal relational density (quantum foam of relations) in which space and time cease to exist, but **couplings** and **nodal fragments** remain intact.

Thus:

- ***Bousso says: "The singularity is inevitable and final."***
- ***RTS responds: "The singularity is inevitable, but temporary."***

Through the subquantum wall, information passes beyond the singularity without violating known laws of physics, because these laws apply only in **open space-time**, not in the quantum foam of relations state.

4. IMPRINTS OF THE GREAT LEAP – OBSERVABLE EVIDENCE

If the universe is cyclic in the RTS sense, then the Great Leap (the transition between cycles) leaves **imprints** in the configuration of the new cycle. These imprints can be searched for in:

- **The cosmic microwave background (CMB)** – anomalies in circular polarization or large-scale ring-like structures, caused by the superposition with the subquantum wall of the previous cycle.
- **The distribution of dark matter** – a preferential clustering on a scale of 10^8 – 10^9 light-years, following a spiral symmetry (as proposed in VG4).
- **Quantum correlations at cosmic distances** – long-lasting entanglement between particles separated by cosmic distances, which cannot be explained except by a shared history in the previous cycle.
- **Low-frequency gravitational waves** – the most promising signal. Fluctuations in the subquantum wall produce extremely low frequencies (below 10^{-9} Hz), which are not yet detectable but are predicted by RTS.

These imprints can be searched for even now, with existing instruments, if we have the patience to listen to the subtle signals of the universe.

5. CONCLUSION – BOUSSO IS RIGHT, BUT RTS GOES FURTHER

Bousso is right within the limits of the standard model. His demonstration is correct and elegant. However, RTS offers a broader framework in which the **subquantum wall** and **nodal fragments** make possible **a richer collection of phenomena** – including cosmic continuity beyond the singularity. Bousso's critique thus becomes an **anchor** for understanding where current physics stops and where our RTS begins.

VG6 is not a rejection of Bousso, but an integration of his work into a broader vision. RTS accepts the singularity but transforms it from an end into a gateway. And this gateway leaves traces that, perhaps, one day we will be able to see.